

Acoustic Measurement & the Anechoic Chamber

How microphones, signal conditioning, and real-time software measure sound, and what an anechoic room reveals

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Course	ME 343-004 Mechanical Lab I, NJIT	Instructor	Prof. B. S. Mani · TA: Saba Mirahsani
Team	Team 4: Luis Ramirez, Samuel Cohen, Edwin Ortiz, Abhin Ghandi, Ramid Brown	Equipment	Calibrated measurement microphone, signal conditioning module, speakers, PC data-acquisition software, anechoic chamber
Format	Demonstration-based lab		

PROJECT SUMMARY

This observational lab, led by Prof. Mani, demonstrated how sound is captured and measured using real acoustic instrumentation. A calibrated microphone fed a signal conditioning module and a computer that plotted sound pressure level in decibels against time, so changes in sound could be seen on screen the moment they were heard. The session highlighted the logarithmic decibel scale, the difference between measured level and perceived loudness, and the striking effect of an anechoic chamber, where sounds that would be loud in a normal room became barely audible.

1. Measurement System

The main sensor was a calibrated microphone connected through a signal conditioning module to a computer running data-acquisition software. The software continuously updated a graph of sound pressure level versus time. As different sound sources were added and changed, the graph responded instantly, making it easy to connect what was heard with what was displayed, even for students new to acoustic measurement.

2. Decibels and Perceived Loudness

Sound played through speakers was captured by the microphone and plotted in real time, showing where the readings matched or differed from what listeners perceived. Sound pressure level is defined as $L_p = 20 \log_{10}(p/p_{ref})$, with $p_{ref} = 20 \mu Pa$. Because the scale is logarithmic, a change that sounds dramatic to the ear can appear as a modest shift on the graph: every 20 dB represents a tenfold change in sound pressure. Seeing this live made the logarithmic nature of the scale far more intuitive than a textbook explanation.

3. The Anechoic Chamber

The most memorable part of the session was the anechoic chamber, a room lined with material designed to absorb sound rather than reflect it. Inside, even sounds that would be loud in a normal room were barely audible, because the sound was absorbed before it could build up. The microphone confirmed this: readings for sources inside the chamber were significantly lower than for the same sources outside.

This showed that the loudness we perceive depends not only on the source but on how sound reflects and reverberates in a room. Remove the reflections, and even a scream seems to vanish. Experiencing that absence of sound in person made the point far more effectively than a lecture or pictures.

4. Discussion

The real-time display was central to the session: rather than only listening, students could observe, analyze, and compare scenarios. Seeing the microphone, signal conditioning hardware, speakers, and software working as one measurement system made acoustic measurement easier to understand than a set of separate instruments. It also linked naturally to earlier labs that used similar data-acquisition setups, since converting a physical phenomenon into an electrical signal and displaying it in real time applies to nearly every type of sensor.

5. Conclusion

The lab gave a clear introduction to the fundamentals of sound measurement: the decibel scale and its logarithmic behavior, real-time graphical display of measurements, the role of microphones and signal conditioning in capturing acoustic data, and the dramatic effect of an anechoic environment on sound levels. Its observational format built a strong intuitive understanding of how sound is measured and what those measurements actually reveal.

References

- Holman, J. P. (2012). *Experimental methods for engineers* (8th ed.). McGraw-Hill.
- Zhu, C. (2009). *ME 343 laboratory introduction*. New Jersey Institute of Technology.